BIOT6900 Assignment 3 Review

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Background

In Module 2, we identified various genes that were potentially relevant to a disease. I chose colon cancer, which is the third most common cancer diagnosed in both men and women in the United States and the second most common cause of cancer deaths.[1] There has been an interesting trend linking colon cancer to increasingly younger populations, which has intensified the search for new drugs to treat the disease.

From Module 2, we carry over the 'targets_ad.csv' file, which is a ranked list of 15 genes which we determined to be the most disease relevant based on our data. (For myself, the CPTAC-COAD and TCGA datasets). In the setup phase, all 15 targets successfully mapped to the proteome in step 1. This indicates that each of the genes which were identified in Module 2 appear in the 'proteome.tsv' file.

In Module 3 we add a second, independent line of evidence to the Module 2 ranking. Each protein is turned into a single vector by a protein language model (in this case ESM C), which is trained by masking amino acids in a sequence and predicting them from the rest, similar to how Large Language Models (LLMs) are trained to predict text. The actual drug information enters later, through labels from Open Targets, which marks a protein as having clinical importance if it is the target of a drug that reached Phase 1, Advanced Clinical or Approved status.[2] We fit a logistic regression on the 960 dimensional ESM-C embeddings of labeled proteins, for both small molecules and antibodies. We then apply this classifier to the 15 targets identified in Module 2. The resulting score judges how similar the selected proteins are to proteins which have actually been targeted by successful drugs. This process uses the protein sequence alone and is irrespective of the underlying disease.

Selecting *w*

The weight *w,* which measures which proportion of the final score comes from the PLM embedding derived prior versus the disease correlation from Module 2. A *w* of 0 represents a total reliance on the Module 3 drug tractability prior, while a *w* of 1 leaves scores unchanged from Module 2. I ended up choosing a value of 0.4, slightly biasing the Module 3 tractability prior over the correlational gene data. I re-generated the list with *w* values from 0.1 to 0.9 to ascertain how the list would shift, and found the 0.4 gives a modest and justifiable shift, without completely disregarding any data. I chose to bias the traceability prior because of the underlying biology of the problem: whether or not a gene is similar to other genes which have been successfully drugged is more informative to the question of drug development. As discussed in Module 2, the biggest caveat of the correlation approach is the ambiguity

over whether or not an identified gene is actually revealing anything about the biology of the disease, or is just correlated with the disease being present. In the case of colon cancer, many of the genes included in the final list are muscle or structural genes, and are not necessarily associated with the cancer itself, but potentially just with the area where the tumor is located. Module 3's tractability prior, however, is informative towards the ultimate end goal of developing drugs for these diseases. This fits with the intention and ordering of the steps: Module 2 identifies a small number of genes which are worthy of investigation, and Module 3 helps narrow that list to a more limited set of genes which may have already developed correlates.

Results

These are the genes, with their Module 2 ranking for reference:

Week 2 rank	Week 3 rank	gene	week2_score	p_sm	p_ab	nearest_drug_targets	final
1	1	FCGBP	0.992	0.091	0.829	ITGA2B (0.95); EPHA1 (0.94); MMP9 (0.94)	0.947
4	2	FLNC	0.984	0.884	0.004	NPEPPS (0.92); PARP3 (0.92); ITK (0.92)	0.92
6	3	KCNMA1	0.976	0.75	0.001	KCNJ3 (0.97); GUCY1A2 (0.96); PDE5A (0.96)	0.787
5	4	CP	0.977	0.405	0.208	ITGB2 (0.94); ATP1B3 (0.94); MMP12 (0.94)	0.773
3	5	ANK2	0.986	0.136	0.002	PDE4D (0.92); RARG (0.92); CACNG3 (0.91)	0.627
7	6	MYLK	0.975	0.2	0.001	TNKS2 (0.94); BRAF (0.93); GPA33 (0.93)	0.627
2	7	MYH11	0.992	0.086	0.001	PSMD11 (0.90); TSG101 (0.90); TOP2B (0.89)	0.587
12	8	MUC2	0.963	0.034	0.37	SEMA4D (0.91); COL15A1 (0.91); GPA33 (0.90)	0.547
9	9	COL1A1	0.968	0.162	0	COL3A1 (0.93); COL1A2 (0.92); SNCA (0.79)	0.507
10	10	COLEC12	0.965	0.011	0.13	ANGPT1 (0.95); LAMC2 (0.92); CD2 (0.92)	0.4
15	11	TNS1	0.96	0.177	0.004	AR (0.95); E2F7 (0.95); CDK13 (0.95)	0.387
8	12	DMD	0.975	0.012	0.001	MAP3K5 (0.93); PDE7A (0.92); ANGPT1 (0.92)	0.307

11	13	EPB41L3	0.964	0.049	0.013	PDE4D (0.90); SNCA (0.86); CACNG3 (0.86)	0.293
14	14	PPP1R12B	0.961	0.012	0.001	HDAC9 (0.95); CACNB2 (0.94); KCNB1 (0.93)	0.173
13	15	SORBS2	0.962	0.005	0.001	KCNB1 (0.94); CACNB2 (0.93); HDAC9 (0.93)	0.12

Of the genes that lost positions, MYH11 had the biggest fall, going from position 2 to position 7. DMD also fell a good bit, going from 8 to 12. The reason for both decline is clear from the data: the traceability prior produced very low scores for *p_sm* and *p_ab* for both MYH11 and DMD. This indicates the prior did not find similar genes that were druggable either in the small particle or antibody classes. Drawing on the real biology, both MYH11 and DMD are muscle genes, so their high Module 2 scores may reflect muscle content differing between normal and tumor samples rather than any actual tumor biology. For genes that gained positions, TNS1 and MUC2 saw the greatest increase, each jumping four positions. However, both of these genes still scored relatively lowly on the tractability prior, indicating these jumps are more likely due to the small absolute differences between the Module 2 scores. As such, it should be noted here that the Module 2 scores were not separated by a large amount, and each of the top 15 showed a strong correlation. Because of this, there is reason to take the ranking difference between Module 2 and 3 with a grain of salt, and to focus more on what the tractability itself score reveals.

Where small molecule and antibody scores diverge

FLNC is the clearest divergence in the list, with *p_sm* of 0.884 and *p_ab* of 0.004. FLNC codes for filamin C, and is intracellular, which explains the low antibody score. The high *p_sm* is somewhat unexpected, considering the underlying protein biology: FLNC has no enzyme activity or small molecule binding site.[3] Considering the nearest drugged targets in the embedding space (NPEPPS, PARP3 and ITK) do not seem to share significant similarities to FLNC, this seems more like an artifact of some aspect of the data processing, rather than any real biology.

Family-split metrics

We checked the prior with out-of-fold predictions, reporting AUROC (how often a random positive is ranked above a random negative) and average precision. With a random split the AUROC was 0.849 and the average precision was 0.733. With a family split, the AUROC fell to 0.748 and the average precision to 0.573. The family split number can probably be trusted more: proteins are very similar within-family, so a large part of the random split score is from grouping the families together.

Anchor variant scores

The variant step is essentially just sanity checking the model. We took four known variants from the assignment's list and looked up a precomputed score for each. The score in this case is judging how surprised the PLM is by a single amino acid change. This score measures how unusual a change is across evolution, not whether it causes disease. For example, an amino acid change that would lead to the early death of an organism 100% of the time is much less tolerated by evolution, so the PLM would be highly surprised to see this mutation. This test only covers swaps of a single amino acid, so they cannot see the potential effects of splicing or non-coding DNA. They are also agnostic of the protein structure or even which cell the protein sits in. So a very negative score means the model finds the change unlikely, not that it is proven harmful.

Limitations and Module 4 targets

The biggest caveat of the tractability estimates is a significant one: the scores are based solely on *historical* druggability data. A gene which is druggable effectively could score 0 if there is no correlated drug already created. Conversely, a gene which is for some reason (binding sites, etc) not druggable, but otherwise has similar residues to other drugs, will score highly. The prior uses sequence only, so it cannot tell whether a protein has a real pocket (which is the purpose of Module 4).

The two targets I will use in Module 4 are KCNMA1 and CP. Module 4 is about predicting physical pockets, and these two are the most interesting in regards to that question. KCNMA1, which ranked third, is an ion channel with a high small molecule score of 0.75, and its nearest drugged proteins are similar potassium channels.. CP, which ranked fourth, has a meaningful score for both small molecules (0.405) and antibodies (0.208), which means that there are more options for potential druggability.

Citations

[1] American Cancer Society. "Colorectal Cancer Statistics." *American Cancer Society*, 14 Jan. 2026, www.cancer.org/cancer/types/colon-rectal-cancer/about/key-statistics.html.

[2] Open Targets Platform documentation. Target tractability. https://platform-docs.opentargets.org/target/tractability

[3] Structure and Function of Filamin C in the Muscle Z-Disc. International Journal of Molecular Sciences 2020;21(8):2696. https://pmc.ncbi.nlm.nih.gov/articles/PMC7216277